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(54) SEMICONDUCTOR PACKAGE HAVING A BALL-BOND INTERCONNECT STRUCTURE AND RELATED METHODS OF MANUFACTURING

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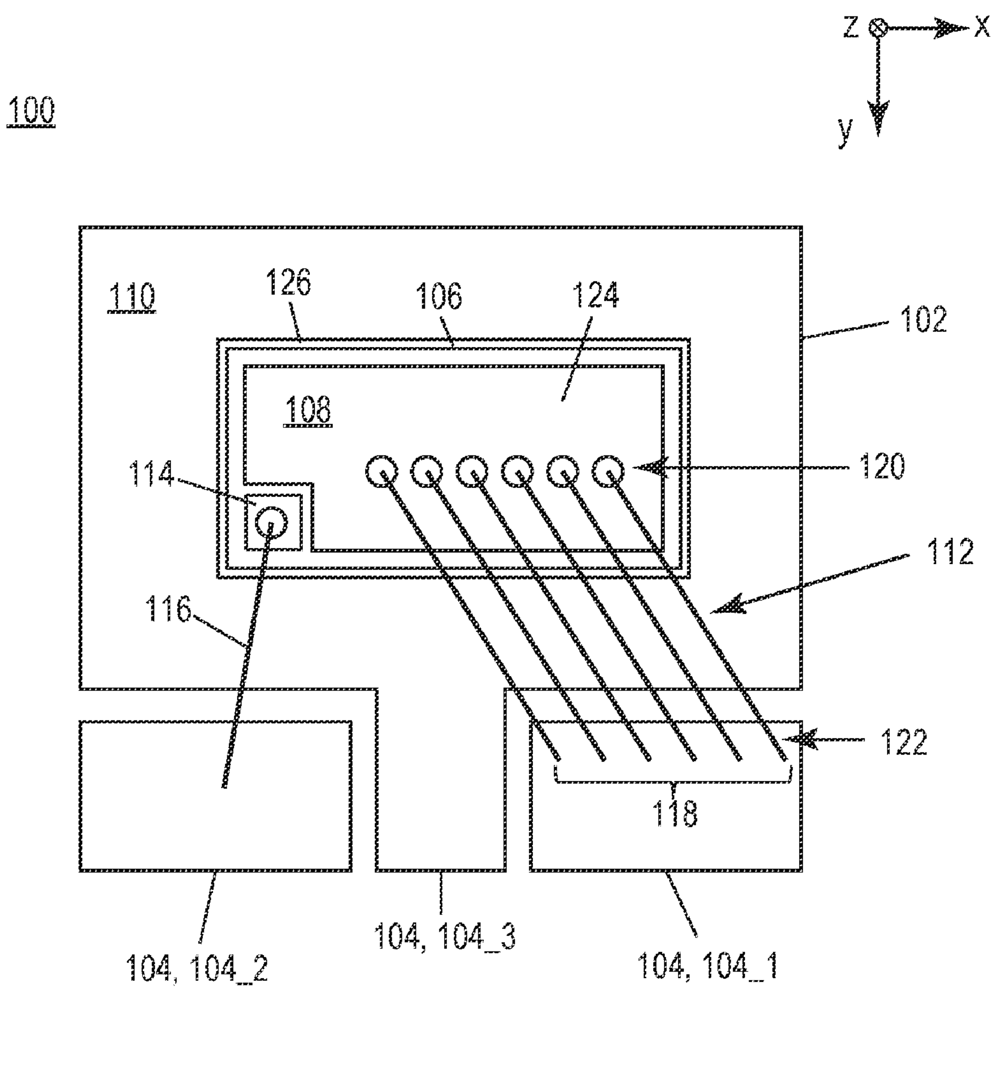
(57) **ABSTRACT**

A semiconductor package includes: a substrate; a plurality of leads; a semiconductor die attached to the substrate at a first side of the semiconductor die, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and a ball-bond interconnect structure connecting the bond pad to a first lead of the plurality of leads. The ball-bond interconnect structure includes at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel. Methods of manufacturing the semiconductor package are also described.



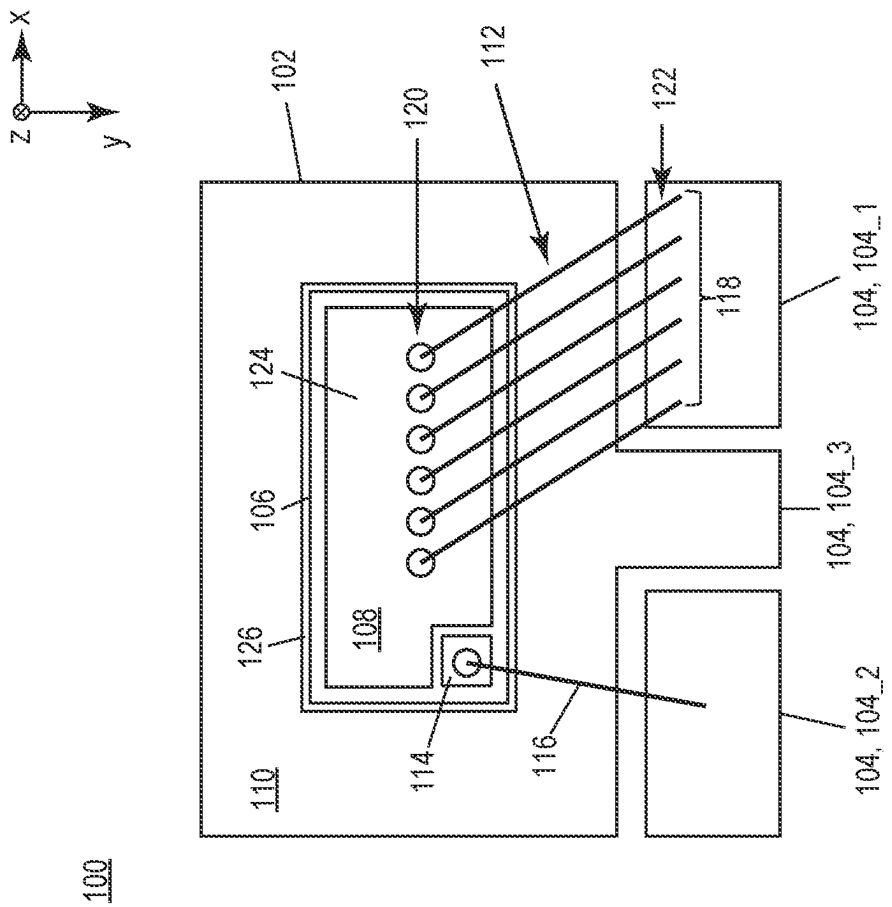

X
Z
y
102
112
120
122
124
106
126
108
110
114
116
118
104, 104_1
104, 104_3
104, 104_2
100

FIG. 1

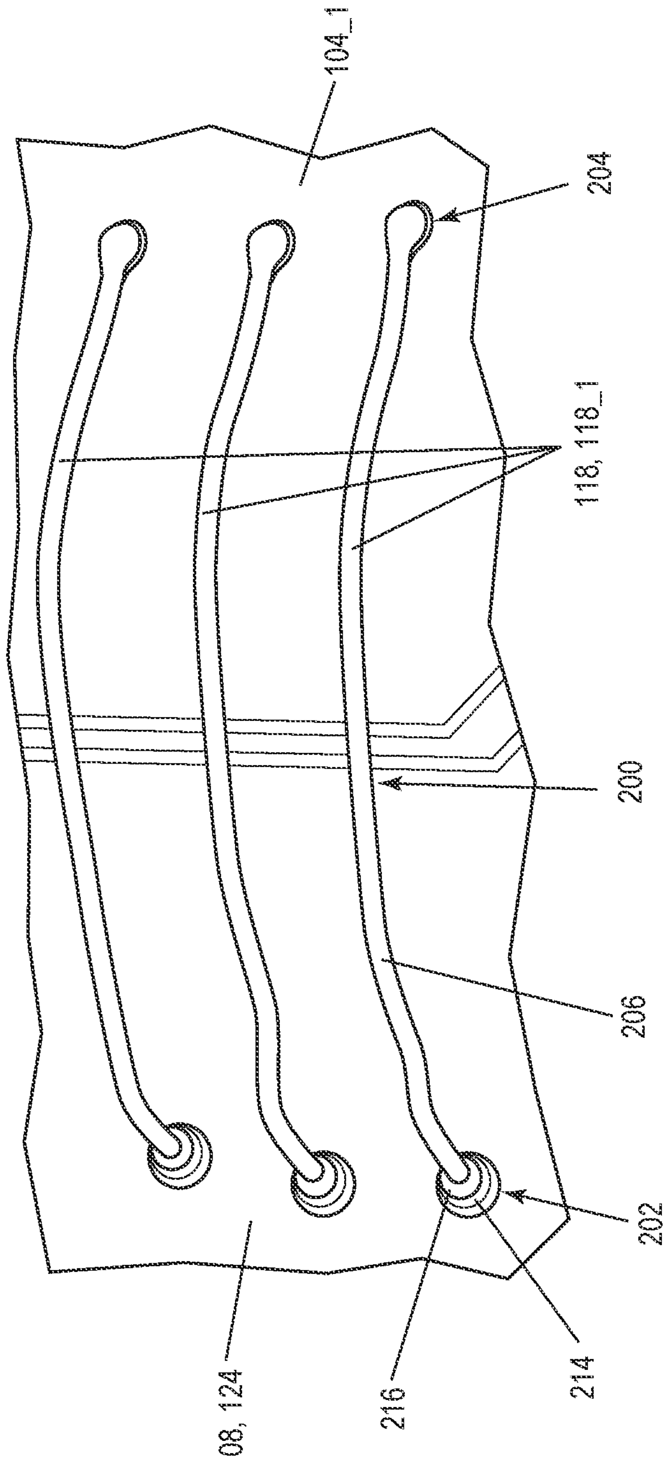


FIG. 2A

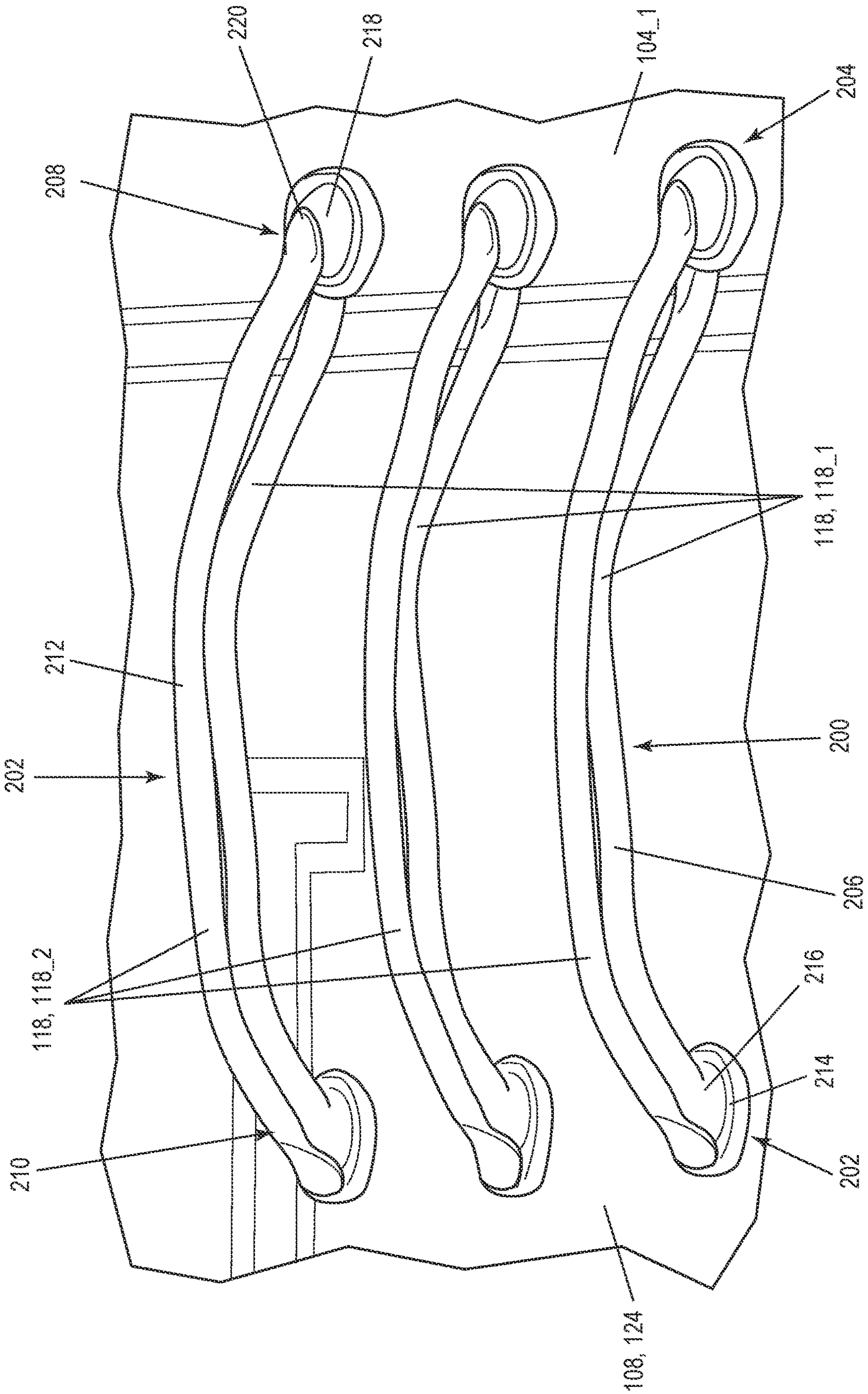

220
218
104_1
204
208
118, 118_1
212
202
200
206
118, 118_2
216
214
210
202
108, 124

FIG. 2B

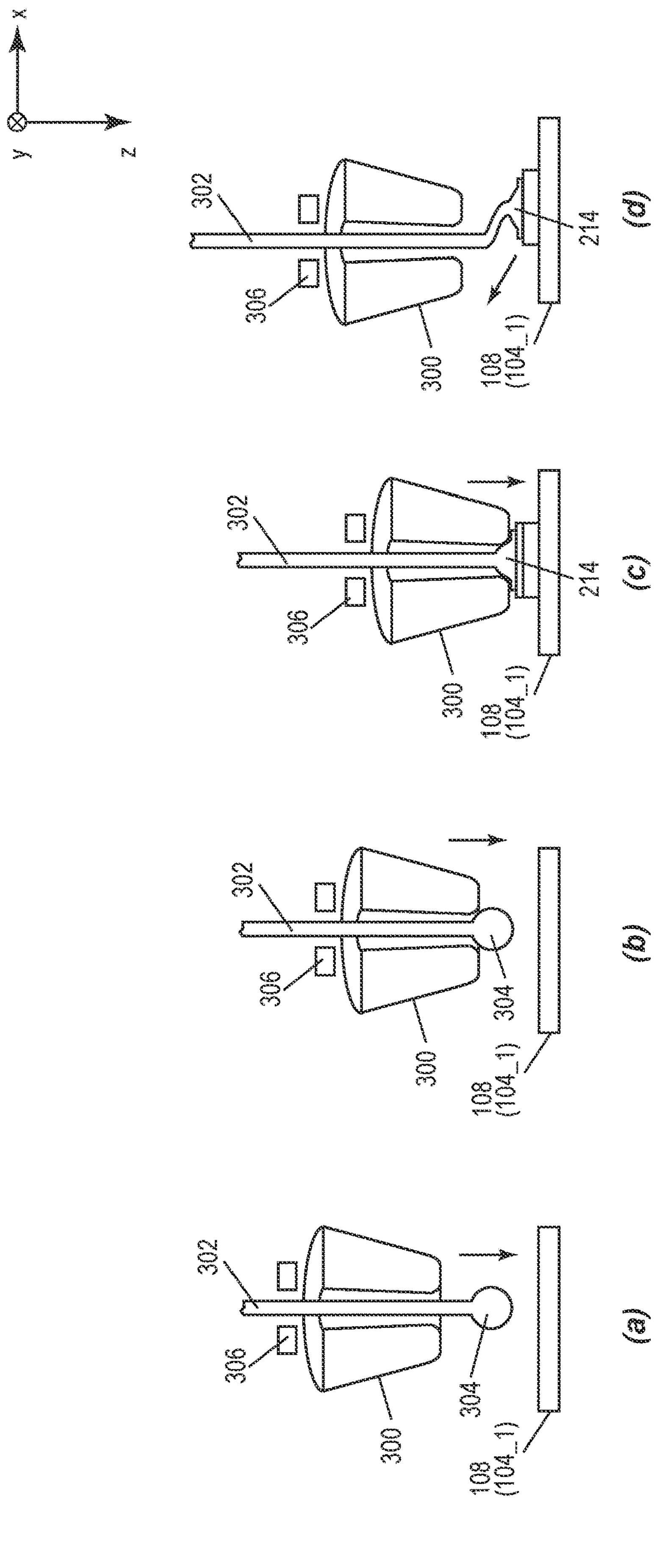

x
y
z
302
306
300
108
(104_1)
214
(d)
302
306
300
108
(104_1)
214
(c)
302
306
300
304
108
(104_1)
(b)
302
306
300
304
108
(104_1)
(a)

FIG. 3

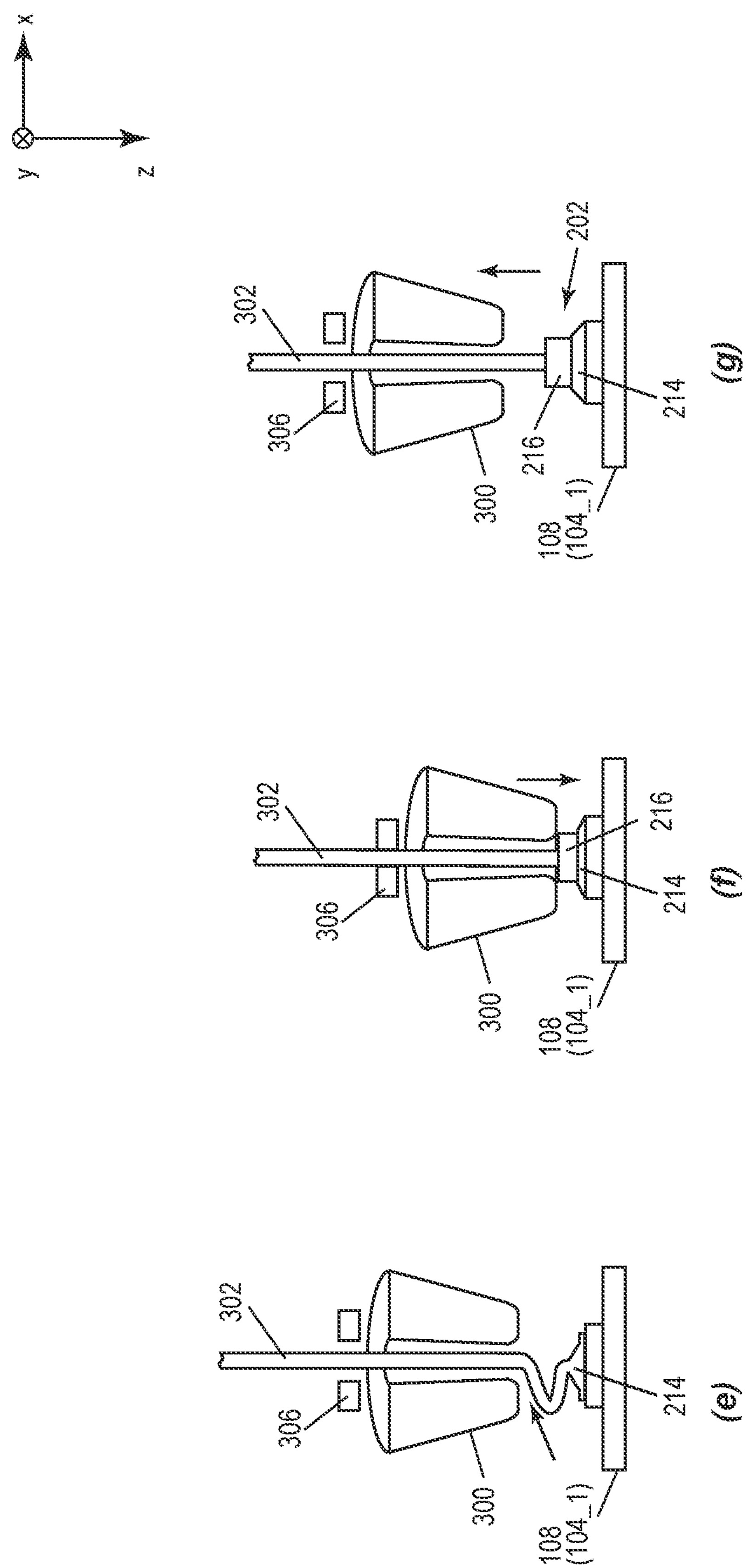

x
y
z
302
306
300
202
216
214
108
(104_1)
(g)
302
306
300
216
214
108
(104_1)
(f)
302
306
300
214
108
(104_1)
(e)

FIG. 3

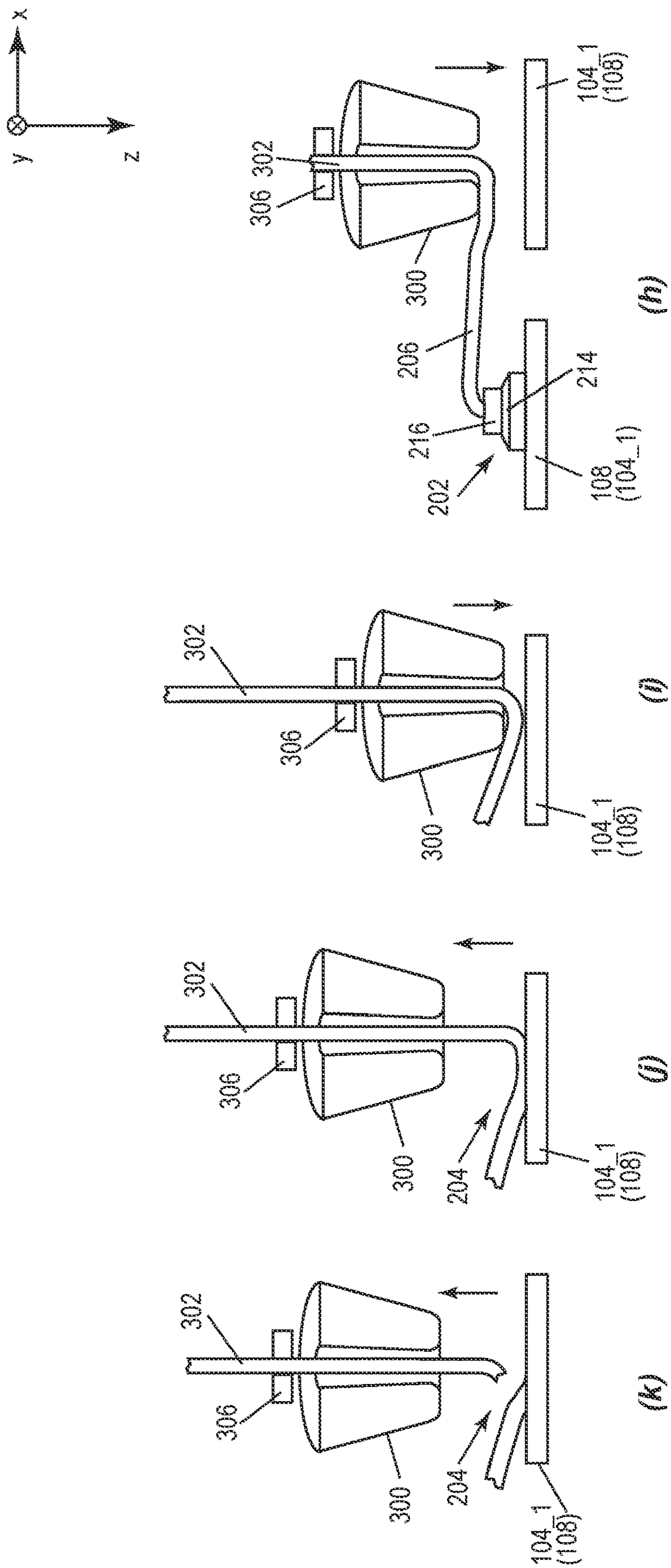

x
y
z
306 302
300
206
216
202
214
108
(104_1)
104_1
(108)
(h)
302
306
300
(i)
302
306
300
204
(j)
302
306
300
204
(k)

FIG. 3

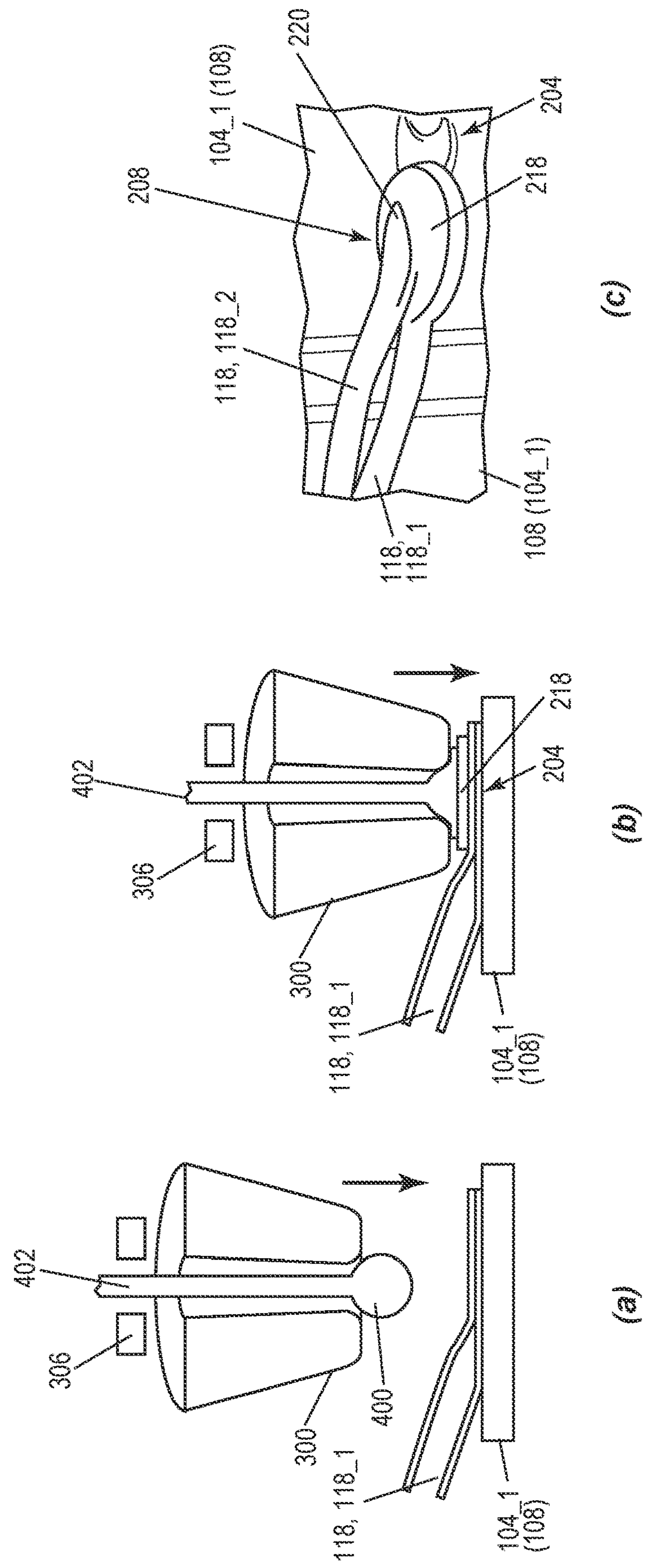

104_1 (108)
220
208
204
218
118, 118_2
118, 118_1
108 (104_1)
(c)
402
306
300
118, 118_1
104_1 (108)
204
218
(b)
402
306
300
400
118, 118_1
104_1 (108)
(a)

FIG. 4

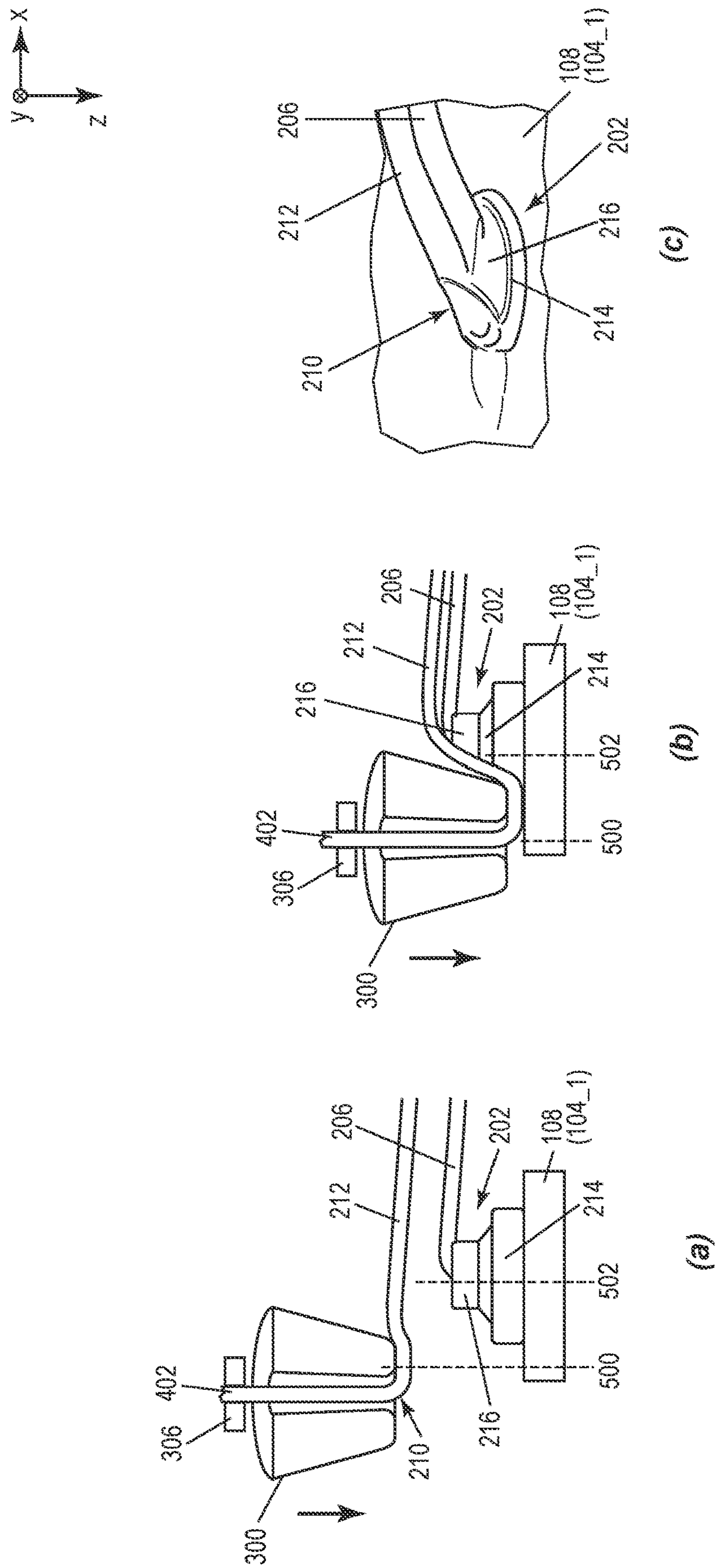


FIG. 5

SEMICONDUCTOR PACKAGE HAVING A BALL-BOND INTERCONNECT STRUCTURE AND RELATED METHODS OF MANUFACTURING

BACKGROUND

[0001] Advancements in SiC (silicon carbide) and GaN (gallium nitride) technologies have yielded smaller bond pad sizes which are up to 30% smaller for higher current density, compared to Si (silicon) technology. Such small bond pads require a high power density interconnect within a small pad/lead area.

[0002] Thus, there is a need for an improved interconnect structure for semiconductor packages with small bond pad sizes.

SUMMARY

[0003] According to an embodiment of a semiconductor package, the semiconductor package comprises: a substrate; a plurality of leads; a semiconductor die attached to the substrate at a first side of the semiconductor die, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and a ball-bond interconnect structure connecting the bond pad to a first lead of the plurality of leads, wherein the ball-bond interconnect structure comprises at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel, wherein the at least two levels of stacked wire bond loops comprises: a first ball-bonded wire loop connected between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and a second ball-bonded wire loop stacked on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

[0004] According to an embodiment of a method of manufacturing a semiconductor package, the method comprises: attaching a first side of a semiconductor die to a substrate, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and forming a ball-bond interconnect structure that connects the bond pad to a first lead, the ball-bond interconnect structure comprising at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel, wherein forming the ball-bond interconnect structure comprises: connecting a first ball-bonded wire loop between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and vertically stacking a second ball-bonded wire loop on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

[0005] Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

BRIEF DESCRIPTION OF THE FIGURES

[0006] The elements of the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts. The features of the various illustrated embodiments can be combined unless they exclude each other. Embodiments are depicted in the drawings and are detailed in the description which follows.

[0007] FIG. **1** illustrates a top plan view of a semiconductor package that includes a multi-level ball-bond interconnect structure for connecting a bond pad of the die to a lead of a package that includes the die.

[0008] FIGS. **2**A and **2**B illustrate respective side perspective views of the multi-level ball-bond interconnect structure during different stages of manufacturing.

[0009] FIG. **3** illustrates an embodiment of forming a ball-bonded wire loop included in the lowermost level of ball-bonded wire loops of the multi-level ball-bond interconnect structure.

[0010] FIG. **4** illustrates an embodiment of attaching a ball end of an upper ball-bonded wire loop to a tail end of a lower ball-bonded wire loop of the multi-level ball-bond interconnect structure.

[0011] FIG. **5** illustrates an embodiment of attaching a bond wire that extends from a stitch of the upper ball-bonded wire loop to a ball end of the lower ball-bonded wire loop.

DETAILED DESCRIPTION

[0012] The embodiments described herein provide a multi-level ball-bond interconnect structure for connecting a bond pad of a semiconductor die (chip) to a lead of a package that includes the semiconductor die. The multi-level ball-bond interconnect structure has at least two levels of ball-bonded wire loops stacked on one another in a direction perpendicular to the die bond pad. The at least two levels of ball-bonded wire loops are attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel. The multi-level ball-bond interconnect structure supports higher current density for smaller bond pad footprints and accommodates a wide range of wire size diameters (e.g., 25 to 75 μm or larger) and materials (e.g., Cu, Au, etc.). The multi-level ball-bond interconnect structure has several advantages over stacked wedge bonding, including higher throughout (e.g., 2×-3× faster units per hour), does not require a thicker bond pad, lower cost (e.g., 3× more cost effective), and wider applicability since ball bonding technology is used in over 90% of semiconductor packaging compared to wedge bonding technology.

[0013] Described next, with reference to the figures, are exemplary embodiments of the multi-level ball-bond interconnect structure and related methods of manufacturing.

[0014] FIG. **1** illustrates a top plan view of a semiconductor package **100**, prior to molding. The semiconductor package **100** includes a substrate **102**, leads **104**, and a semiconductor die **106** attached to the substrate **102** at a back side of the semiconductor die **106**. The semiconductor die **106** has a first bond pad **108** at the front side of the semiconductor die **106** which is opposite the back side.

[0015] The semiconductor die **106** may be a logic die such as a processor die, memory die, etc., a power semiconductor die such as a power transistor die, a power diode die, a half bridge die, etc., or a die that combines logic and power devices on the same semiconductor substrate. In one embodiment, the semiconductor die **106** is a vertical semiconductor die having a primary current path between the bond pad **108** at the die front side and a bond pad (out of view) at the die back side. Examples of vertical power semiconductor dies include but are not limited to power Si MOSFETs (metal-oxide-semiconductor field-effect transistors), IGBTs (insulated-gate bipolar transistors), SiC MOSFETs, GaN HEMTs (high-electron mobility transistors), etc.

[0016] More than one semiconductor die **106** may be attached to the same or different substrate **102** such that the semiconductor package **100** may include one or more semiconductor dies **106**. For example, the semiconductor package **100** may be a discrete in that the package **100** includes a single semiconductor die **106** such as a low-side or high-side power transistor die. In another example, the semiconductor package **100** may include one or more low-side power transistor dies in parallel and one or more high-side power transistor dies in parallel to form a half bridge. A gate driver (not shown) for each semiconductor die **106** also may be included in the semiconductor package **100**.

[0017] The substrate **102** to which at least one semiconductor die **106** is attached may be a lead frame. For example, the semiconductor die **106** may be attached to a die paddle **110** of the lead frame and the package leads **104** may be part of the lead frame. The lead frame features (e.g., leads **104**, die paddle **110**, etc.) may be defined by etching, stamping, etc. of a metallic sheet such as a Cu (copper) sheet.

[0018] The semiconductor package **100** also includes a multi-level ball-bond interconnect structure **112** that connects the first bond pad **108** of the semiconductor die **106** to a first lead **104_1** of the package leads **104**. For example, in the case of a vertical power transistor die, the first bond pad **108** may be a source pad and the multi-level ball-bond interconnect structure **112** may connect the source pad **108** to a source lead **104_1** of the package. Further in the case of a vertical power transistor die, a gate pad **114** may be disposed at the same side of the semiconductor die **106** as the source pad **108** and may be electrically connected to another lead **104_2**, e.g., by at least bond wire **116**. In the case of a lead frame as the substrate **102**, the die paddle **110** may form one of the package leads **104_3**.

[0019] The ball-bond interconnect structure **112** is multi-level in that the ball-bond interconnect structure **112** includes at least two levels of ball-bonded wire loops **118** stacked on one another in a direction (z direction in FIG. **1**) perpendicular to the front side of the semiconductor die **106**. The at least two levels of ball-bonded wire loops **118** are attached to one another at opposite ends **120**, **122** of the stacked ball-bonded wire loops **118** such that the at least two levels of stacked ball-bonded wire loops **118** are electrically in parallel to support higher current density. The multi-level ball-bond interconnect structure **112** may be particularly advantageous for smaller bond pad footprints having a relatively high current density, e.g., in the case of the semiconductor die **106** being a SiC or GaN die. In the case of a SiC or GaN, the semiconductor die **106** may have an area less than 2 mm^2 and/or a current density greater than 4.5 A/mm^2 and/or be attached to the substrate **102** by lead-free solder **126**. The multi-level ball-bond interconnect structure **112** also may be used with larger bond pad footprints, e.g., in the case of the semiconductor die **106** being made from Si.

[0020] FIGS. **2**A and **2**B illustrate respective side perspective views of the multi-level ball-bond interconnect structure **112** during different stages of manufacturing. FIG. **2**A shows the multi-level ball-bond interconnect structure **112** after a first (lowermost) level **200** of ball-bonded wire loops **118** is formed, and FIG. **2**B shows the ball-bond interconnect structure **112** after a second level **202** of ball-bonded wire loops **118** is vertically stacked on the lowermost level **200** of ball-bonded wire loops **118**.

[0021] The lowermost level **200** of ball-bonded wire loops **118** includes a plurality of first ball-bonded wire loops **118_1** connected in parallel between the die bond pad **108** and the corresponding package lead **104_1**. Three (3) first ball-bonded wire loops **118_1** are visible in the partial views of FIGS. **2**A and **2**B. However, the lowermost level **200** of ball-bonded wire loops **118** may include two (2), three (3) or more first ball-bonded wire loops **118_1** connected in parallel between the die bond pad **108** and the corresponding package lead **104_1**. Each first ball-bonded wire loop **118_1** has a ball end **202** attached to one of the die bond pad **108** or the package lead **104_1**, a tail (stitch) end **204** attached to the other one of the die bond pad **108** or package lead **104_1**, and a wire section **216** that is compressed into the ball end **202** and flattened.

[0022] The second level **202** of ball-bonded wire loops **118** includes a plurality of second ball-bonded wire loops **118_2** stacked on the plurality of first ball-bonded wire loops **118_1**. Three (3) second ball-bonded wire loops **118_2** are visible in the partial view of FIG. **2**B. However, the second level **202** of ball-bonded wire loops **118** may include two (2), three (3) or more second ball-bonded wire loops **118_2** stacked on the plurality of first ball-bonded wire loops **118_1**. In one embodiment, the second level **202** of ball-bonded wire loops **118** includes the same number of ball-bonded wire loops **118** as the first level **200** of ball-bonded wire loops **118**. However, the second level **202** of ball-bonded wire loops **118** may include fewer ball-bonded wire loops **118** than the first level **200** of ball-bonded wire loops **118**, e.g., one-half, one-third, etc.

[0023] Each second ball-bonded wire loop **118_2** has a ball end **208** attached to the tail end **204** of a corresponding one of the first ball-bonded wire loops **118_1** and a tail end **210** attached to the ball end **202** of the corresponding first ball-bonded wire loop **118_1**. Each second ball-bonded wire loop **118_2** also has a wire section **212** that is folded in a lateral (e.g., x) direction and extends between the ball end **210** and the tail end **208** of the second ball-bonded wire loop **118_2**.

[0024] As shown in FIG. **2**A, the ball end **202** of each first ball-bonded wire loop **118_1** has a ball bump **214** attached to one of the die bond pad **108** or the package lead **104_1** and a wire section **216** that is compressed into the ball bump **214** and flattened as part of a wire folding process. As shown in FIG. **2**B, the ball end **208** of each second ball-bonded wire loop **118_2** has a ball bump **218** attached to the tail end **204** of the corresponding first ball-bonded wire loop **118_1** and a stitch **220** on the ball bump **218** of the second ball-bonded wire loop **118_2**. The ball bump **218** of each second ball-bonded wire loop **118_2** is first bonded to the tail end **204** of the corresponding first ball-bonded wire loop **118_1**, and then the stitch **220** is formed on top of the ball bump **218** as part of a wire folding process. The tail end **210** of each second ball-bonded wire loop **118_2** is attached to the stitch **216** and the ball bump **214** of the corresponding first ball-bonded wire loop **118_1**.

[0025] In one embodiment, the outermost layer **124** of the die bond pad **108** to which the multi-level ball-bond interconnect structure **112** is attached comprises copper and the plurality of first ball-bonded wire loops **118_1** and the plurality of second ball-bonded wire loops **118_2** each comprise copper. In another embodiment, the outermost layer **124** of the die bond pad **108** to which the multi-level ball-bond interconnect structure **112** is attached comprises gold and the plurality of first ball-bonded wire loops **118_1** and the plurality of second ball-bonded wire loops **118_2** each comprise gold. Other monometallic systems or a bimetallic system may be used.

[0026] Separately or in combination, the plurality of first ball-bonded wire loops **118_1** and the plurality of second ball-bonded wire loops **118_2** each have a diameter in a range of 25 μm to 75 μm. Separately or in combination, the plurality of first ball-bonded wire loops **118_1** has a first diameter and the plurality of second ball-bonded wire loops **118_2** has a second diameter different than the first diameter such that the plurality of first ball-bonded wire loops **118_1** may be thinner or thicker than the plurality of second ball-bonded wire loops **118_2**.

[0027] Two (2) levels **200**, **202** of stacked ball-bonded wire loops **118** are shown in FIG. 2B. However, this is just an example. The multi-level ball-bond interconnect structure **112** may have more than two (2) levels of ball-bonded wire loops **118** stacked on one another in the vertical direction, where each overlying level of ball-bonded wire loops **118** runs in the opposite direction as the underlying level of ball-bonded wire loops **118** such that interleaved stacks of ball and tail end bonds are formed at both ends of the multi-level ball-bond interconnect structure **112**. For example, a plurality of third ball-bonded wire loops (not shown in FIGS. 2A and 2B) may be stacked on the plurality of second ball-bonded wire loops **118_2**, where each third ball-bonded wire loop has a ball end attached to the tail end **210** of a corresponding one of the second ball-bonded wire loops **118_2**, a tail end attached to the ball end **208** of the corresponding second ball-bonded wire loop **118_2**, and a wire section that is compressed into the ball end of the wire loop and flattened.

[0028] FIG. **3** illustrates an embodiment of forming one of the first ball-bonded wire loop **118_1** included in the lowermost level **200** of ball-bonded wire loops **118**. The process illustrated in FIG. **3** may be repeated one or more times to form two (2) or more first ball-bonded wire loops **118_1** that are connected in parallel between two (2) bond sites.

[0029] In step (a) of FIG. **3**, an electrical flame off (EFO) process is performed using a capillary **300** threaded with a bond wire **302** to form a free air ball **304** at the end of the bond wire **302**.

[0030] In step (b) of FIG. **3**, the capillary **300** captures the free air ball **304** with the capillary clamp **306** released and moves in the z direction to the first bond site which may be the die bond pad **108** or the corresponding package lead **104_1**.

[0031] In step (c) of FIG. **3**, force and/or ultrasonic energy are applied to form the ball bump **214** of a first ball-bonded wire loop **118_1** at the first bond site such that the ball bump **214** is attached to one of the die bond pad **106** or the corresponding package lead **104_1**. The second bond site is the other one of the die bond pad **106** or the package lead **104_1** and to which the ball bump **214** is not attached.

[0032] Steps (d) and (e) of FIG. **3** involve a wire folding process during which the capillary **300** is moved in the z direction away from the first bond site and changes lateral (e.g., x or y) direction, as indicated by the angled arrows, to fold the bond wire **302**.

[0033] In step (f) of FIG. **3**, the clamp **306** closes and the capillary **300** moves downwards in the z direction towards the first bond site and compresses the folded section in the bond wire **302** (e.g. z or s-shaped fold) onto the previously formed ball bump **214**. This folding and compressing creates a flattened area that allows for the stacking of multiple wire loops while mitigating bonding related issues. For example, the stitch of a second ball-bonded wire loop stacked on the first ball-bonded wire loop is likely to be malformed if the capillary is directly pressed onto the rounded ball bump without the creation of a flattened bonding area over the ball bump **304** using the folding and compressing steps.

[0034] In step (g) of FIG. **3**, the capillary **300** moves in the z direction away from the first bond site with the clamp **306** released as part of a wire looping process. The flattening of the wire loop also enables having a lower loop angle and accordingly making the stacking of multiple wire loop layers possible. The wire looping process further involves moving the capillary **300** with the clamp **306** released in a lateral (e.g., x) direction to the second bond site, folding the bond wire **302** in the lateral direction and defining the length of a folded wire section **206** that extends between the ball end **202** and the tail end **204** of the first ball-bonded wire loop **118_1**.

[0035] In step (h) of FIG. **3**, the clamp **306** is engaged and the capillary **300** moves in the z direction to the second bond site.

[0036] In step (i) of FIG. **3**, the clamp **306** is released and force and/or ultrasonic energy are applied to attach the bond wire **302** that extends from the stitch **216** to the second bond site.

[0037] In step (j) of FIG. **3**, the capillary **300** moves away from the second bond site in the z direction with the clamp **306** released.

[0038] In step (k) of FIG. **3**, the clamp **306** is engaged and the capillary **300** continues to move away from the second bond site in the z direction, severing the bond wire **302** to form the tail (stitch) end **204** of the first ball-bonded wire loop **118_1** and complete the first ball-bonded wire loop **118_1**.

[0039] FIG. **4** illustrates an embodiment of attaching the ball end **208** of a second ball-bonded wire loop **118_2** to the tail end **204** of one of the first ball-bonded wire loops **118_1**. The process illustrated in FIG. **4** may be repeated for each first ball-bonded wire loop **118_1** in the lowermost level **200** of ball-bonded wire loops **118**. The process illustrated in FIG. **4** may be repeated for each additional level of stacked ball-bonded wire loops **118**, but changing direction for each additional level such that the ball end of each new ball-bonded wire loop **118** being formed is bonded to the tail end of the previously formed ball-bonded wire loop **118**.

[0040] In step (a) of FIG. **4**, a free air ball **400** has already been formed at the end of a bond wire **402** fed through the capillary **300** and the capillary **300** moves in the z direction with the clamp **306** released towards the tail end **204** of the corresponding first ball-bonded wire loop **118_1**. The free air ball **400** may be formed as explained above in connection with step (a) of FIG. **3**.

[0041] In step (b) of FIG. **4**, force and/or ultrasonic energy are applied to form the ball bump **218** of the second ball-bonded wire loop **118_2**. The ball bump **218** of the second ball-bonded wire loop **118_2** is attached to the tail end **204** of the corresponding first ball-bonded wire loop **118_1**.

[0042] In step (c) of FIG. **4**, the bond wire **402** is folded in a lateral (e.g., x) direction and force and/or ultrasonic energy are applied to form the stitch **220** on the ball bump **218** of the second ball-bonded wire loop **118_2**, e.g., as explained above in connection with steps (d) through (f) of FIG. **3**. The capillary **300** is then moved as part of a wire looping process that involves vertical and lateral movement to bend or loop the bond wire **402** to the second bond site, e.g., as explained above in connection with steps (g) and (h) of FIG. **3**.

[0043] FIG. **5** illustrates an embodiment of attaching the bond wire **402** that extends from the stitch **220** of the second ball-bonded wire loop **118_2** to the ball end **202** of the corresponding first ball-bonded wire loop **118_1**. The process illustrated in FIG. **5** may be repeated for each first ball-bonded wire loop **118_1** in the lowermost level **200** of ball-bonded wire loops **118**. The process illustrated in FIG. **5** may be repeated for each additional level of stacked ball-bonded wire loops **118**, but changing direction for each additional level such that the tail end of each ball-bonded wire loop **118** being formed is bonded to the ball end of the underlying ball-bonded wire loop **118**.

[0044] In step (a) of FIG. **5**, the capillary **300** is moved such that in a longitudinal (lengthwise) direction (x direction in FIG. **5**) of the ball-bond interconnect structure **112**, a center point **500** of the tail end **210** of the second ball-bonded wire loop **118_2** is off-centered in the longitudinal direction from the center point **504** of the ball bump **214** of the corresponding first ball-bonded wire loop **118_1**. Such longitudinal off-centering allows for a more planar stacking at the ends **202**, **210** of the stacked wire loops **118_1**, **118_2**. The positioning of the capillary **300** also defines the length of the folded wire section **212** that extends between the ball end **208** and the tail end **210** of the second ball-bonded wire loop **118_2**.

[0045] In step (b) of FIG. **5**, the capillary clamp **306** is released and force and/or ultrasonic energy are applied to attach the bond wire **402** that extends from the stitch **220** of the second ball-bonded wire loop **118_2** to the ball end **202** of the corresponding first ball-bonded wire loop **118_1**.

[0046] In step (c) of FIG. **5**, the bond wire **402** is severed to form the tail end **210** of the second ball-bonded wire loop **118_2** and complete the second ball-bonded wire loop **118_2**. The tail end **210** of the second ball-bonded wire loop **118_2** is attached to the ball end **202** of the corresponding first ball-bonded wire loop **118_2**.

[0047] The process illustrated in FIGS. **4** and **5** may be repeated for each additional level of ball-bonded wire loops **118** that form the ball-bond interconnect structure **112**, by stacking a desired number of ball-bonded wire loops **118** on the next lowermost level of ball-bonded wire loops **118**. Each ball-bonded wire loop **118** in the newest level of ball-bonded wire loops **118** has a ball end attached to the tail end of the corresponding ball-bonded wire loop **118** in the next lowermost level of ball-bonded wire loops **118**, a tail end attached to the ball end of the corresponding ball-bonded wire loop, and a wire section that is compressed into the ball end of the new ball-bonded wire loop and flattened.

[0048] Although the present disclosure is not so limited, the following numbered examples demonstrate one or more aspects of the disclosure.

[0049] Example 1. A semiconductor package, comprising: a substrate; a plurality of leads; a semiconductor die attached to the substrate at a first side of the semiconductor die, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and a ball-bond interconnect structure connecting the bond pad to a first lead of the plurality of leads, wherein the ball-bond interconnect structure comprises at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel, wherein the at least two levels of stacked wire bond loops comprises: a first ball-bonded wire loop connected between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and a second ball-bonded wire loop stacked on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

[0050] Example 2. The semiconductor package of example 1, wherein the ball end of the first ball-bonded wire loop comprises a ball bump attached to one of the bond pad or the first lead and a stitch on the ball bump, wherein the ball end of the second ball-bonded wire loop comprises a ball bump attached to the tail end of the first ball-bonded wire loop and a stitch on the ball bump of the second ball-bonded wire loop, and wherein the tail end of the second ball-bonded wire loop is attached to the stitch and the ball bump of the first ball-bonded wire loop.

[0051] Example 3. The semiconductor package of example 1 or 2, wherein the at least two levels of stacked wire bond loops comprises: a third ball-bonded wire loop stacked on the second ball-bonded wire loop, the third ball-bonded wire loop comprising a ball end attached to the tail end of the second ball-bonded wire loop, a tail end attached to the ball end of the second ball-bonded wire loop, and a wire section that is compressed into the ball end of the third ball-bonded wire loop and flattened.

[0052] Example 4. The semiconductor package of any of examples 1 through 3, wherein the semiconductor die is a SiC or GaN die.

[0053] Example 5. The semiconductor package of example 4, wherein an outermost layer of the bond pad comprises copper, and wherein the first ball-bonded wire loop and the second ball-bonded wire loop both comprise copper.

[0054] Example 6. The semiconductor package of example 5, wherein the first ball-bonded wire loop and the second ball-bonded wire loop both have a diameter in a range of 25 μm to 75 μm.

[0055] Example 7. The semiconductor package of example 5 or 6, wherein the first ball-bonded wire loop has a first diameter, and wherein the second ball-bonded wire loop has a second diameter different than the first diameter.

[0056] Example 8. The semiconductor package of any of examples 5 through 7, wherein the semiconductor die is attached to the substrate by lead-free solder.

[0057] Example 9. The semiconductor package of any of examples 4 through 8, wherein the semiconductor die has an area less than 2 mm^2.

[0058] Example 10. The semiconductor package of example 9, wherein the semiconductor die has a current density greater than 4.5 A/mm^2.

[0059] Example 11. A method of manufacturing a semiconductor package, the method comprising: attaching a first side of a semiconductor die to a substrate, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and forming a ball-bond interconnect structure that connects the bond pad to a first lead, the ball-bond interconnect structure comprising at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel, wherein forming the ball-bond interconnect structure comprises: connecting a first ball-bonded wire loop between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and vertically stacking a second ball-bonded wire loop on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

[0060] Example 12. The method of example 11, wherein connecting the first ball-bonded wire loop between the bond pad and the first lead comprises: attaching a first ball bump to one of the bond pad or the first lead; forming a first stitch on the first ball bump; attaching a first wire that extends from the first stitch to the other one of the bond pad or the first lead; and severing the first wire to form the tail end of the first ball-bonded wire loop, wherein vertically stacking the second ball-bonded wire loop on the first ball-bonded wire loop comprises: attaching a second ball bump to the tail end of the first ball-bonded wire loop; forming a second stitch on the second ball bump; attaching a second wire that extends from the second stitch to the ball end of the first ball-bonded wire loop; and severing the second wire to form the tail end of the second ball-bonded wire loop.

[0061] Example 13. The method of example 12, wherein attaching the second wire that extends from the second stitch of the second ball-bonded wire loop to the ball end of the first ball-bonded wire loop comprises: in a longitudinal direction of the ball-bond interconnect structure, offsetting a center point of the tail end of the second ball-bonded wire loop from a center point of the first ball bump of the first ball-bonded wire loop.

[0062] Example 14. The method of any of examples 11 through 13, wherein forming the ball-bond interconnect structure comprises: stacking a third ball-bonded wire loop on the second ball-bonded wire loop, the third ball-bonded wire loop comprising a ball end attached to the tail end of the second ball-bonded wire loop, a tail end attached to the ball end of the second ball-bonded wire loop, and a wire section that is compressed into the ball end of the third ball-bonded wire loop and flattened.

[0063] Example 15. The method of any of examples 11 through 14, wherein the semiconductor die is a SiC or GaN die.

[0064] Example 16. The method of example 15, wherein an outermost layer of the bond pad comprises copper, and wherein the first ball-bonded wire loop and the second ball-bonded wire loop both comprise copper.

[0065] Example 17. The method of example 16, wherein the first ball-bonded wire loop and the second ball-bonded wire loop both have a diameter in a range of 25 μm to 75 μm.

[0066] Example 18. The method of any of examples 15 through 17, wherein attaching the first side of the semiconductor die to the substrate comprises: soldering the first side of the semiconductor die to the substrate using lead-free solder.

[0067] Terms such as "first", "second", and the like, are used to describe various elements, regions, sections, etc. and are also not intended to be limiting. Like terms refer to like elements throughout the description.

[0068] As used herein, the terms "having", "containing", "including", "comprising" and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles "a", "an" and "the" are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

[0069] The expression "and/or" should be interpreted to include all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression "A and/or B" should be interpreted to mean only A, only B, or both A and B. The expression "at least one of" should be interpreted in the same manner as "and/or", unless expressly noted otherwise. For example, the expression "at least one of A and B" should be interpreted to mean only A, only B, or both A and B.

[0070] It is to be understood that the features of the various embodiments described herein may be combined with each other, unless specifically noted otherwise.

[0071] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

What is claimed is:

1. A semiconductor package, comprising:

a substrate;

a plurality of leads;

a semiconductor die attached to the substrate at a first side of the semiconductor die, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side; and

a ball-bond interconnect structure connecting the bond pad to a first lead of the plurality of leads,

wherein the ball-bond interconnect structure comprises at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel,

wherein the at least two levels of stacked wire bond loops comprises:

a first ball-bonded wire loop connected between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and

a second ball-bonded wire loop stacked on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

2. The semiconductor package of claim **1**,

wherein the ball end the first ball-bonded wire loop comprises a ball bump attached to one of the bond pad or the first lead and a stitch on the ball bump,

wherein the ball end of the second ball-bonded wire loop comprises a ball bump attached to the tail end of the first ball-bonded wire loop and a stitch on the ball bump of the second ball-bonded wire loop, and

wherein the tail end of the second ball-bonded wire loop is attached to the stitch and the ball bump of the first ball-bonded wire loop.

3. The semiconductor package of claim **1**, wherein the at least two levels of stacked wire bond loops comprises:

a third ball-bonded wire loop stacked on the second ball-bonded wire loop, the third ball-bonded wire loop comprising a ball end attached to the tail end of the second ball-bonded wire loop, a tail end attached to the ball end of the second ball-bonded wire loop, and a wire section that is compressed into the ball end of the third ball-bonded wire loop and flattened.

4. The semiconductor package of claim **1**, wherein the semiconductor die is a SiC or GaN die.

5. The semiconductor package of claim **4**, wherein an outermost layer of the bond pad comprises copper, and wherein the first ball-bonded wire loop and the second ball-bonded wire loop both comprise copper.

6. The semiconductor package of claim **5**, wherein the first ball-bonded wire loop and the second ball-bonded wire loop both have a diameter in a range of 25 μm to 75 μm.

7. The semiconductor package of claim **5**, wherein the first ball-bonded wire loop has a first diameter, and wherein the second ball-bonded wire loop has a second diameter different than the first diameter.

8. The semiconductor package of claim **5**, wherein the semiconductor die is attached to the substrate by lead-free solder.

9. The semiconductor package of claim **4**, wherein the semiconductor die has an area less than 2 mm^2.

10. The semiconductor package of claim **9**, wherein the semiconductor die has a current density greater than 4.5 A/mm^2.

11. A method of manufacturing a semiconductor package, the method comprising:

attaching a first side of a semiconductor die to a substrate, the semiconductor die having a bond pad at a second side of the semiconductor die opposite the first side;

forming a ball-bond interconnect structure that connects the bond pad to a first lead, the ball-bond interconnect structure comprising at least two levels of ball-bonded wire loops stacked on one another, and attached to one another at opposite ends of the stacked ball-bonded wire loops such that the at least two levels of stacked ball-bonded wire loops are electrically in parallel,

wherein forming the ball-bond interconnect structure comprises:

connecting a first ball-bonded wire loop between the bond pad and the first lead, the first ball-bonded wire loop comprising a ball end attached to one of the bond pad or the first lead, a tail end attached to the other one of the bond pad or the first lead, and a wire section that is compressed into the ball end of the first ball-bonded wire loop and flattened; and

vertically stacking a second ball-bonded wire loop on the first ball-bonded wire loop, the second ball-bonded wire loop comprising a ball end attached to the tail end of the first ball-bonded wire loop, a tail end attached to the ball end of the first ball-bonded wire loop, and a wire section that is compressed into the ball end of the second ball-bonded wire loop and flattened.

12. The method of claim **11**,

wherein connecting the first ball-bonded wire loop between the bond pad and the first lead comprises:

attaching a first ball bump to one of the bond pad or the first lead;

forming a first stitch on the first ball bump;

attaching a first wire that extends from the first stitch to the other one of the bond pad or the first lead; and

severing the first wire to form the tail end of the first ball-bonded wire loop,

wherein vertically stacking the second ball-bonded wire loop on the first ball-bonded wire loop comprises:

attaching a second ball bump to the tail end of the first ball-bonded wire loop;

forming a second stitch on the second ball bump;

attaching a second wire that extends from the second stitch to the ball end of the first ball-bonded wire loop; and

severing the second wire to form the tail end of the second ball-bonded wire loop.

13. The method of claim **12**, wherein attaching the second wire that extends from the second stitch of the second ball-bonded wire loop to the ball end of the first ball-bonded wire loop comprises:

in a longitudinal direction of the ball-bond interconnect structure, offsetting a center point of the tail end of the second ball-bonded wire loop from a center point of the first ball bump of the first ball-bonded wire loop.

14. The method of claim **11**, wherein forming the ball-bond interconnect structure comprises:

stacking a third ball-bonded wire loop on the second ball-bonded wire loop, the third ball-bonded wire loop comprising a ball end attached to the tail end of the second ball-bonded wire loop, a tail end attached to the ball end of the second ball-bonded wire loop, and a wire section that is compressed into the ball end of the third ball-bonded wire loop and flattened.

15. The method of claim **11**, wherein the semiconductor die is a SiC or GaN die.

16. The method of claim **15**, wherein an outermost layer of the bond pad comprises copper, and wherein the first ball-bonded wire loop and the second ball-bonded wire loop both comprise copper.

17. The method of claim **16**, wherein the first ball-bonded wire loop and the second ball-bonded wire loop both have a diameter in a range of 25 μm to 75 μm.

18. The method of claim **15**, wherein attaching the first side of the semiconductor die to the substrate comprises:

soldering the first side of the semiconductor die to the substrate using lead-free solder.

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